

Octet

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July 7, 2026

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Octet



Octet

An eight-lane keyboard rhythm game with a built-in beat-mapping editor and an online map community.

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Octet is a fast, keyboard-native rhythm game. Notes fall down eight lanes mapped to the home row, and you hit them the instant they cross the judgment line. Drop in any song and the built-in editor finds the tempo and the beats for you, so mapping is placement, not hand-syncing by ear. Browse and download community-made maps from the shared Map Hub.

Octet is the successor to QWERTY — same core mechanic, rebuilt from the ground up.

Features

- **Eight-lane play** on the home row — `A S D F / J K L ;` — with taps, chords, and holds, tiered timing windows, combo, live accuracy, and letter grades.
- **Built-in editor** with automatic BPM and beat detection, a waveform + beat-grid overlay, snap-to-grid placement, multiple difficulties, and playtest-in-editor.
- **Map Hub** — browse, search, and download community-made maps, played straight from the client. Publishing today is a manual, repo-owner step (see Maps below); per-map and global leaderboards are planned but not live.
- **Tight sync** via an audio-driven conductor clock and a per-machine calibration screen for audio and input latency.
- **Windows and macOS** native builds.

Controls

Lane	1	2	3	4	5	6	7	8
Key	A	S	D	F	J	K	L	;

All keys are rebindable. Scroll speed, offsets, and accessibility options are in settings.

Download

Native builds for Windows and macOS are published under Releases.

[!NOTE] No release build has been exported yet — gameplay, editor, and Map Hub are all playable today from source (see Build from source below), but packaging hasn't happened. Watch Releases for the first build.

Build from source

Requires **Godot 4.7**. A Firebase project is only needed for the (not-yet-built) online accounts layer — see Tech below.

```
# clone, then open in Godot 4.7
godot4 project.godot
```

Firebase configuration (web config keys) goes in a local config that is git-ignored; the service account key stays server-side in Cloud Functions and is never committed. See `docs/PROJECT_BRIEF.md` §6.4.

Maps

Maps are `.octet` bundles — a zip of the audio, one or more `.oct` chart files, and optional cover/background art. The `.oct` chart is JSON: metadata, timing points, and a list of notes. Full format in `docs/PROJECT_BRIEF.md` §4 and `docs/CODEMAPS/data.md`.

Community maps in **Map Hub** are served today as plain files from this repo's `maps/` directory over `raw.githubusercontent.com` (see `docs/MAP_HUB_PUBLISHING.md`) — no backend required. Per-map and global leaderboards are not live yet; the UI marks them “coming soon” rather than faking scores.

Roadmap

- **M1** — First playable: full eight-lane gameplay, judgments, calibration, results.
- **M2** — Editor: waveform, manual BPM/offset, snapped placement, difficulties, save/export.
- **M3** — Automatic BPM + beat detection.
- **M4** — Online: accounts, map hub, downloads, per-map leaderboards.
- **M5** — Community, global ranks, polish, installers.

Documentation

- `ARCHITECTURE.md` — how the systems fit together and why
- `CONTRIBUTING.md` — dev setup, running tests, project conventions
- `CHANGELOG.md` — release history
- `docs/PROJECT_BRIEF.md` — full design and technical spec
- `docs/DESIGN_BRIEF.md` — visual identity and screens
- `docs/MAP_HUB_PUBLISHING.md` — Map Hub manifest schema and publishing
- `docs/CODEMAPS/` — architecture, gameplay, editor, UI, data, and dependency codemaps

Tech

Godot 4.7 (GDScript — no GDExtension in use; the audio-analysis FFT is hand-rolled in pure GDScript). Map Hub runs today as plain files served from this repo over `raw.githubusercontent.com`. Firebase (Auth, Firestore, Cloud Storage, Cloud Functions) is the planned backend for accounts and leaderboards (Roadmap M4) and is not yet wired in.

Changelog

All notable changes to Octet are documented in this file.

The format is based on Keep a Changelog (keepachangelog.com/en/1.1.0).

[1.0.0] — 2026-07-07

First playable release. Eight-lane rhythm gameplay, a full beat-mapping editor, and a working (file-hosted) map community, built on Godot 4.7.

ADDED

- **Eight-lane play** on the home row (`A S D F` / `J K L ;`) with taps, chords, and holds, tiered timing windows, combo, live accuracy, and letter grades (SS/S/A/B/C/D).
- **Audio-driven Conductor clock** — timing derived from playback position, corrected for output latency and per-machine calibration, never frame delta.
- **Calibration screen** — tap-to-the-beat routine measures and stores audio/input offset so judging matches what the player actually hears and feels.
- **Built-in beat-mapping editor** — waveform view, snap-to-grid beat grid, note placement/selection, undo/redo, multiple difficulties, and playtest-in-editor.
- **Automatic BPM and beat detection** — a hand-rolled FFT (`editor/dsp_fft.gd`) and onset/tempo analysis (`editor/audio_analysis.gd`) seed BPM/offset on audio import, so mapping starts from placement, not hand-syncing by ear.
- `.oct` **chart format** and `.octet` **bundles** — JSON charts (metadata, timing points, notes) packaged with audio and cover art into a distributable zip.
- **Six bundled songs** across 2-3 difficulties each, plus cover art support.
- **Map Hub community screen** — browse, download, and play community maps, served today as files from this repo over `raw.githubusercontent.com` (no backend required).
- **Settings screen** — rebindable keys, scroll speed, accessibility options, audio buses (Music/SFX), all persisted per machine via `SettingsStore`.
- **Player profile screen** and **local best-score tracking** per chart.
- **Windows and macOS** native builds via Godot 4.7 (GL Compatibility renderer).

- Full UI fidelity pass — fonts, rendering, and mockup rebuilds across every screen, matched against the Claude Design mockups (main menu, song select, gameplay HUD, results, calibration, editor).

KNOWN LIMITATIONS

- **Online accounts and global leaderboards are not live** — Firebase (Auth, Firestore, Storage) is scaffolded but stubbed (`Net.is_online()` always false). Per-map and global leaderboards are UI-stubbed as “coming soon,” not simulated. This is Roadmap milestone M4.